Evolution Of Mobile Communications

LEC:2

Sub: **mobile communication systems**
Fourth class
BY: **MOHAMMED IHSAN HASHIM**

Reference
Wireless Communications: Principles and Practice,Theodore S. Rappaport

First Generation (1G) systems

- **Representative Standards**

1. Advanced Mobile Phone Service (AMPS):

1. used in South America, Australia, and China.
2. Two **25-MHz** bands are allocated to AMPS.
3. Each of these bands is split in two operators each operator is allocated only **12.5** MHz in each direction
4. The channels are **spaced 30 kHz**, which allows a **total of 416** channels per operator.
5. **21** channels are allocated for **control**, leaving **395 to carry calls**.
6. The control channels are data channels operating at **10 kbps**.
7. The conversation channels carry the conversations in **analog using FM.**

Second Generation (2G) systems

1. developed in early 1990s to provide higher quality signals
2. higher data rates for support of digital services,
3. greater capacity.
4. Digital voice coding and modulation
5. Security (Encryption).
6. Error detection and correction.
7. Multiple channels per cell.

Representative Standards:

	GSM	IS-95
Year introduced	1990	1993
Developed in	Europe	North America
BS transmission band	935~960 MHz	869~894 MHz
MS transmission band	890~915 MHz	824~849 MHz
Spacing between forward and reverse channels	45 MHz	45 MHz
Channel bandwidth	200 KHz	1250 KHz
No. of duplex channels	125	20
Mobile unit maximum power	20 W	0.2 W
Users per channel	8	35
Modulation	GMSK	QPSK
Carrier bit rate	270.8 kbps	9.6 kbps

the 2.5G system

The main features of the 2.5G are:

1. These new standards represent 2.5G technology and allow existing 2G equipment to be **modified and upgrades** the software to support higher data rate
2. Support **web browsing** and receive **e-mail message using (GPRS) stanard**
3. Speed: 64-144 kbps
4. Built-in camera can be included in the mobile station (Camera phones).

General Packet Radio Service (GPRS)

1. GPRS is **packet switched** technology
2. **based on GSM** cellular network infrastructure.
3. **Speed: up to 114 kbps**
4. The GPRS allows multiple slots of a GSM radio channel be dedicated to an individual user **(Physical channel can be shared between different mobile users).**
5. Physical channel is only assigned when data needs to be transmitted or received.
6. The GPRS gives GSM subscribers access to **e-mail** and **internet** using their mobile phone.

the 3G systems

- **the main features of the 3G systems are:**
- developed in the early 2000s,
- **High transmission rate and the support of multimedia services:** Multiple-megabit internet services, video calls, and mobile TV using a single mobile device
- **Data rate:** around 2Mbps

3G enhancements

- 1. Enhanced audio and video streaming
- 2. Several times higher data speed
- 3. Video-conferencing support
- 4. Web and WAP browsing at higher speeds
- 5. IPTV (TV through the Internet) support

Fourth Generation (4G) systems

The main features of the 4G systems are

1. provides **broadband Internet** access, expanded multimedia services
2. Allow users to download a full-length film within five minutes and able to stream **UHD tv.**
3. Higher speed **0.1~1 Gbps**.
4. **More security.**
5. **Higher capacity.**
6. **Lower cost** than previous generations.
7. Provides digital system with voice over-IP (VOIP) technology.
8. **IPv6 Core**.
9. support **packet switching** instead of circuit switching in voice and video calls
10. **OFDM** is used instead of CDMA.

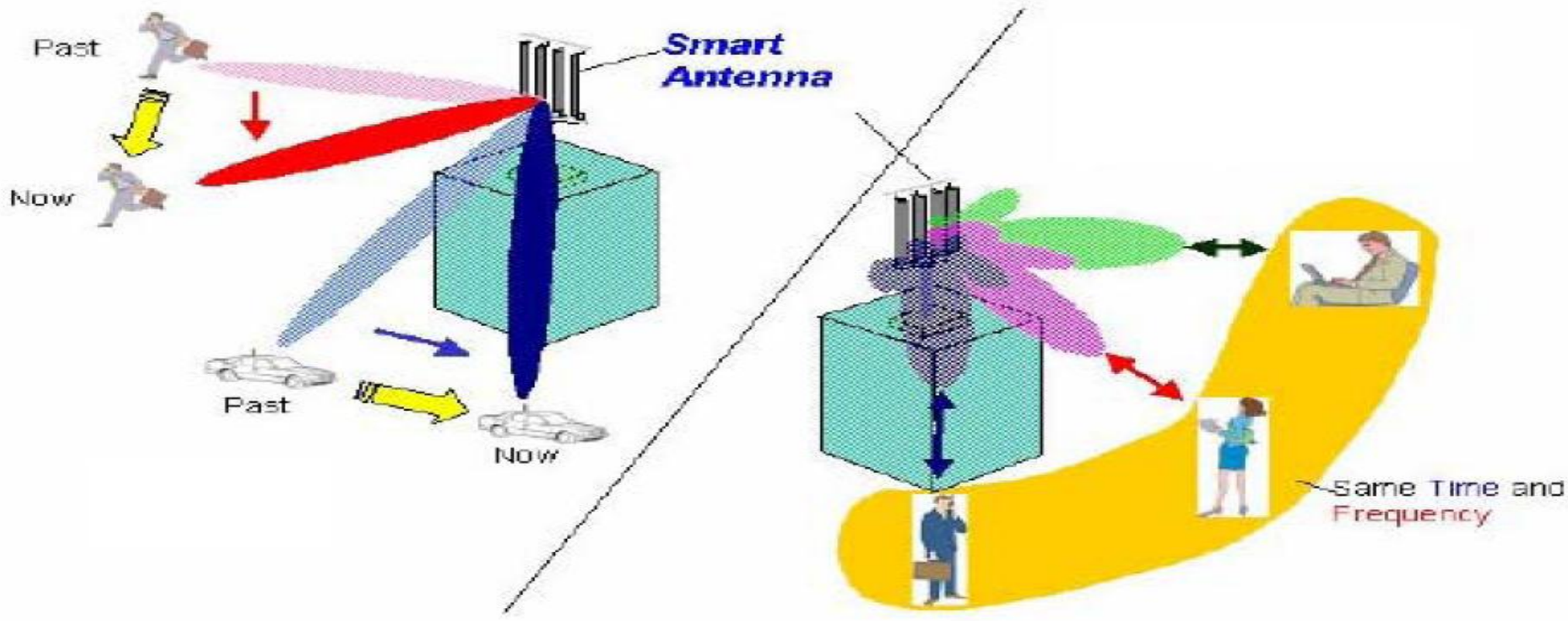


Technologies used in 4G

1. Smart Antenas
2. IPv6
3. VoIP
4. OFDM
5. Multiple-input and multiple-output (MIMO)

Smart Antennas

- Beam radio signals directly at a user to follow the user as they move
  - Allow the same radio frequency to be used for other users without worry of interference

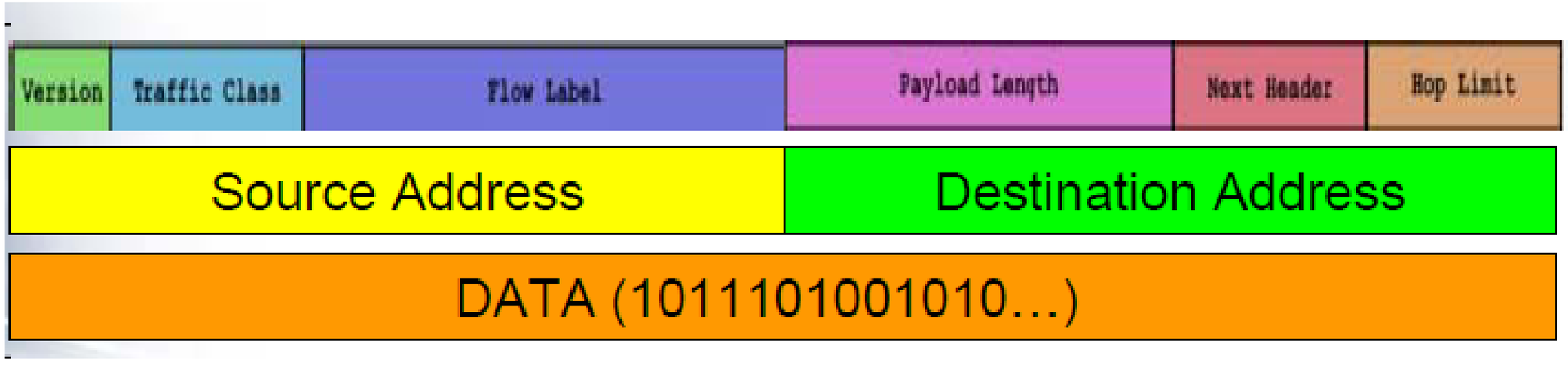


Smart Antenna Technology

IPv6

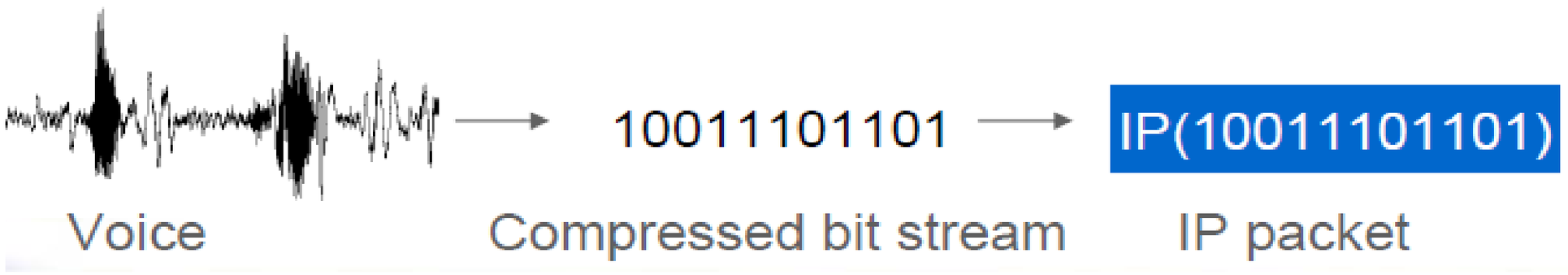
-> Mobile IPv6 have been proposed to reduce the handoff Mobile latency and the number of lost packets

- The Internet Protocol (IP) is the method or protocol which data is sent from one computer to another on the internet.
- It includes128 bits, which is 4 times more than



Mobile VoIP

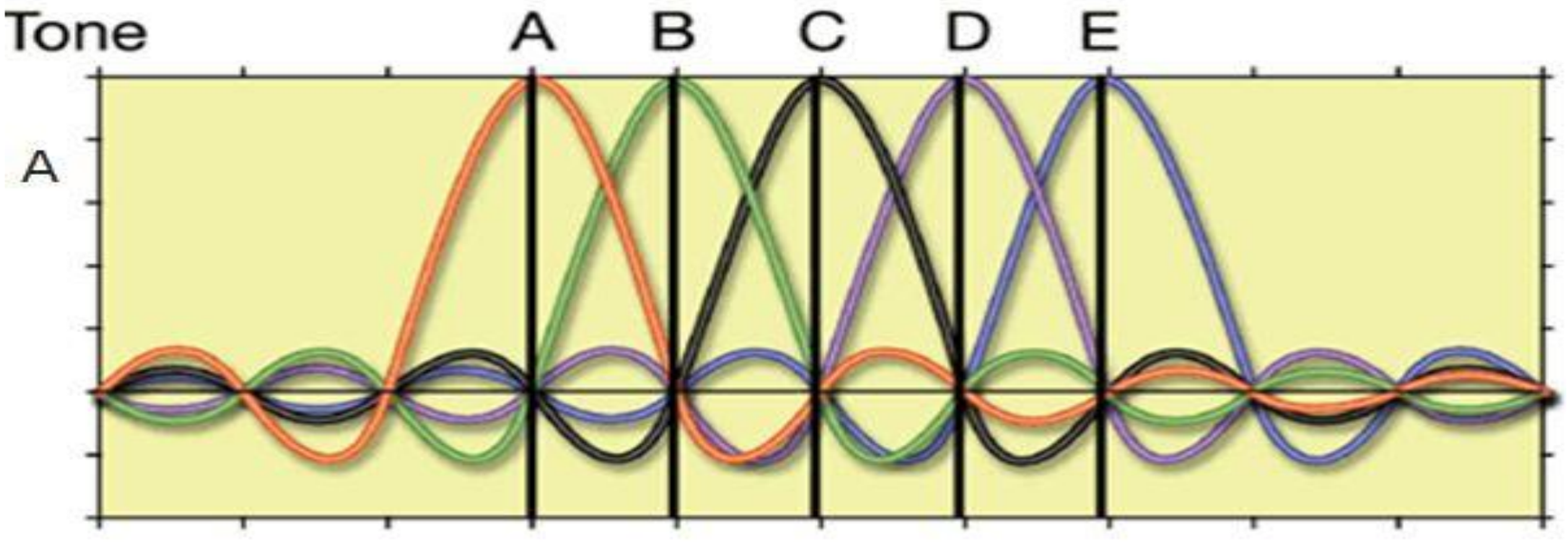
- "Voice Over Internet Protocol"
- Allows only packets (IP) to be transferred eliminating complexity of 2 protocols over the same circuit
- All voice data will be wrapped up in a packet
  - Lower latency data transmission (faster transmission)
- Samples voice between 8,000 & 64,000 times per second and creates stream of bits which is then compressed and put into a packet.
- Increases battery life due to greater data compression



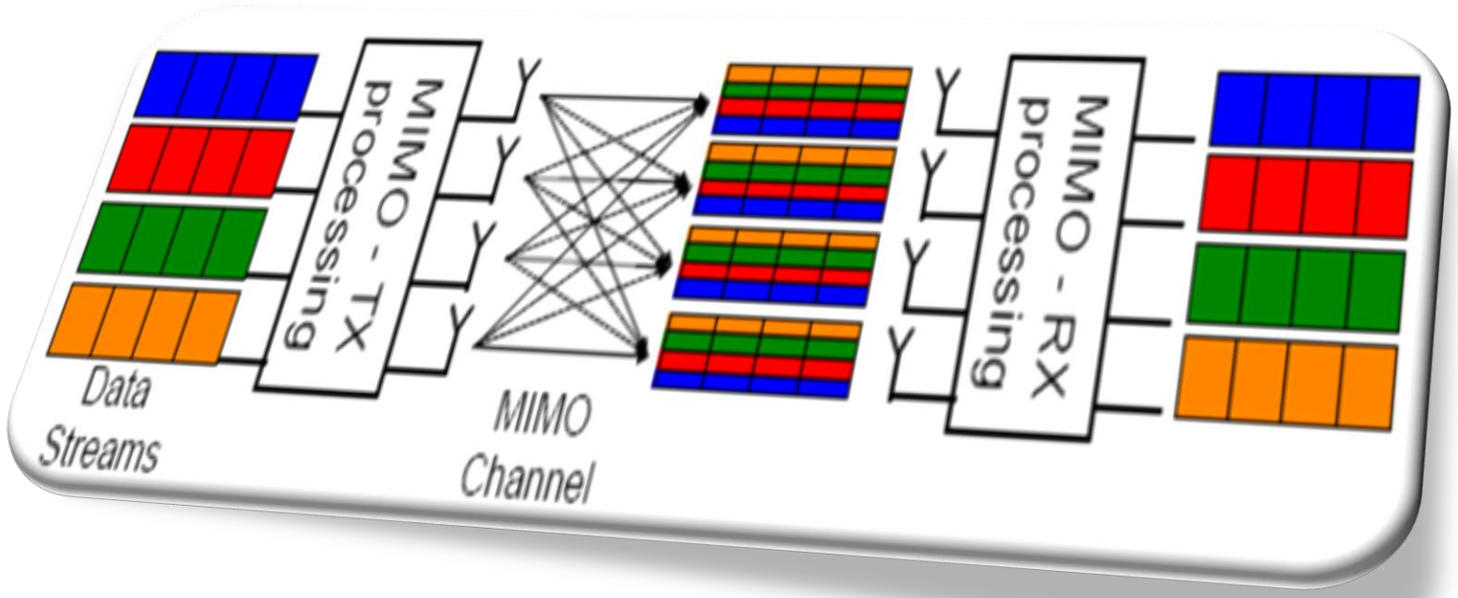
OFDM

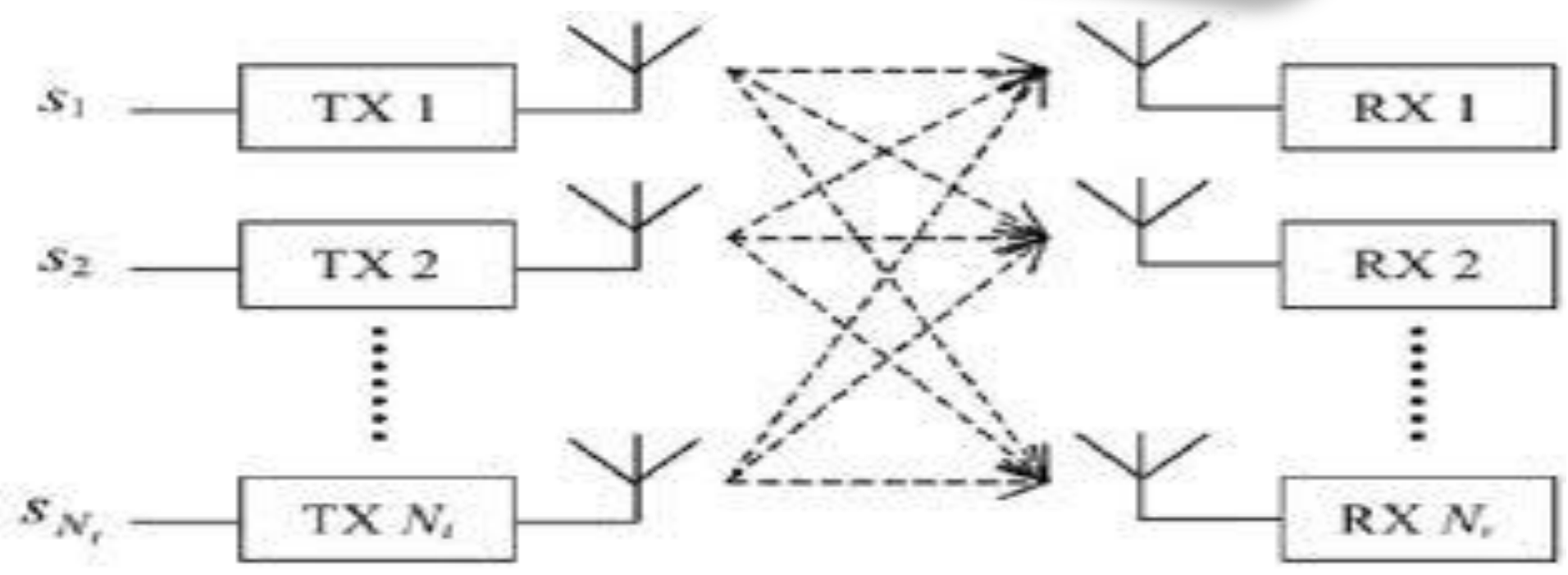
- Orthogonal Frequency Division Multiplexing
- Allows for transfer of more data than other forms of multiplexing (time, frequency, code, etc)
- Allows for use of almost the entire frequency band

  -No gaps to prevent interference needed
- Currently used in WiMax(802.16) and WiFi(802.11a/g)
- The frequencies are spaced so that the signals do not interfere with each other (no cross talk)



Multiple-input and multiple-output (MIMO)

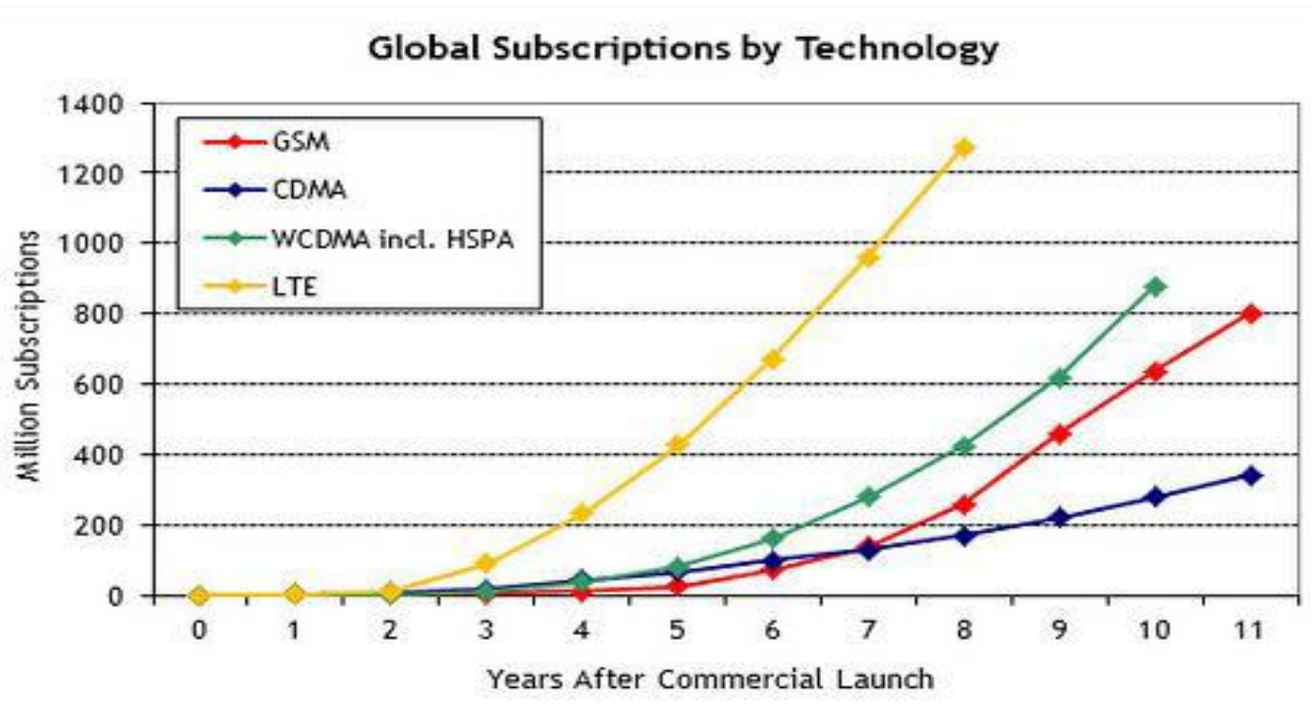




LTE Advanced (LTE-A)

- **The main features of the LTE-Advanced are:**

1. enhancement of the Long Term Evolution (LTE) standard
2. higher capacity,
3. increased peak data rate **3 Gbps Downlink** & 1.5 Gbps Uplink
4. Increased number of simultaneously active subscribers.
5. support up to 8x8 MIMO and 128 QAM in downlink direction.
6. LTE-Advanced is now available **in 31 countries** (including South Korea, Australia, France, Germany, U.K. and the U.S.).



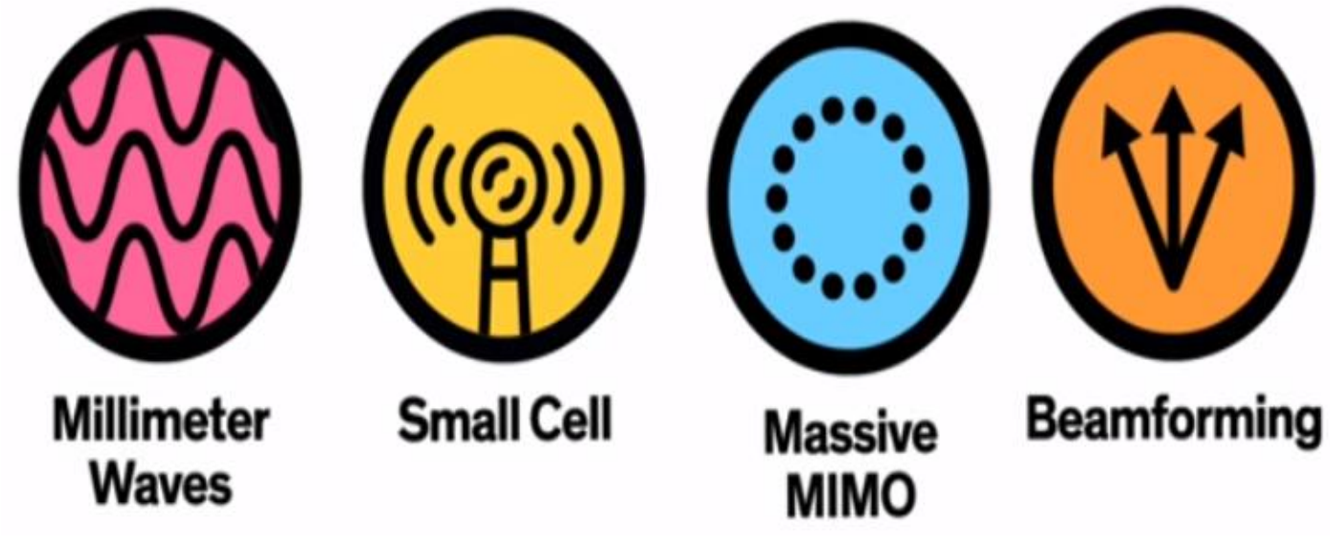
5G Enabling Techniques

By
MOHAMMED I. HABELALMATEEN

Introduction

There are four keys common inventions that will enable the evolution of cellular networks to support 5G and serve future users demands:

1. **Millimeter wave (mm-Waves)**
2. **HetNets using small cell technology**
3. **Beam-forming technique**
4. **Massive MIMO**

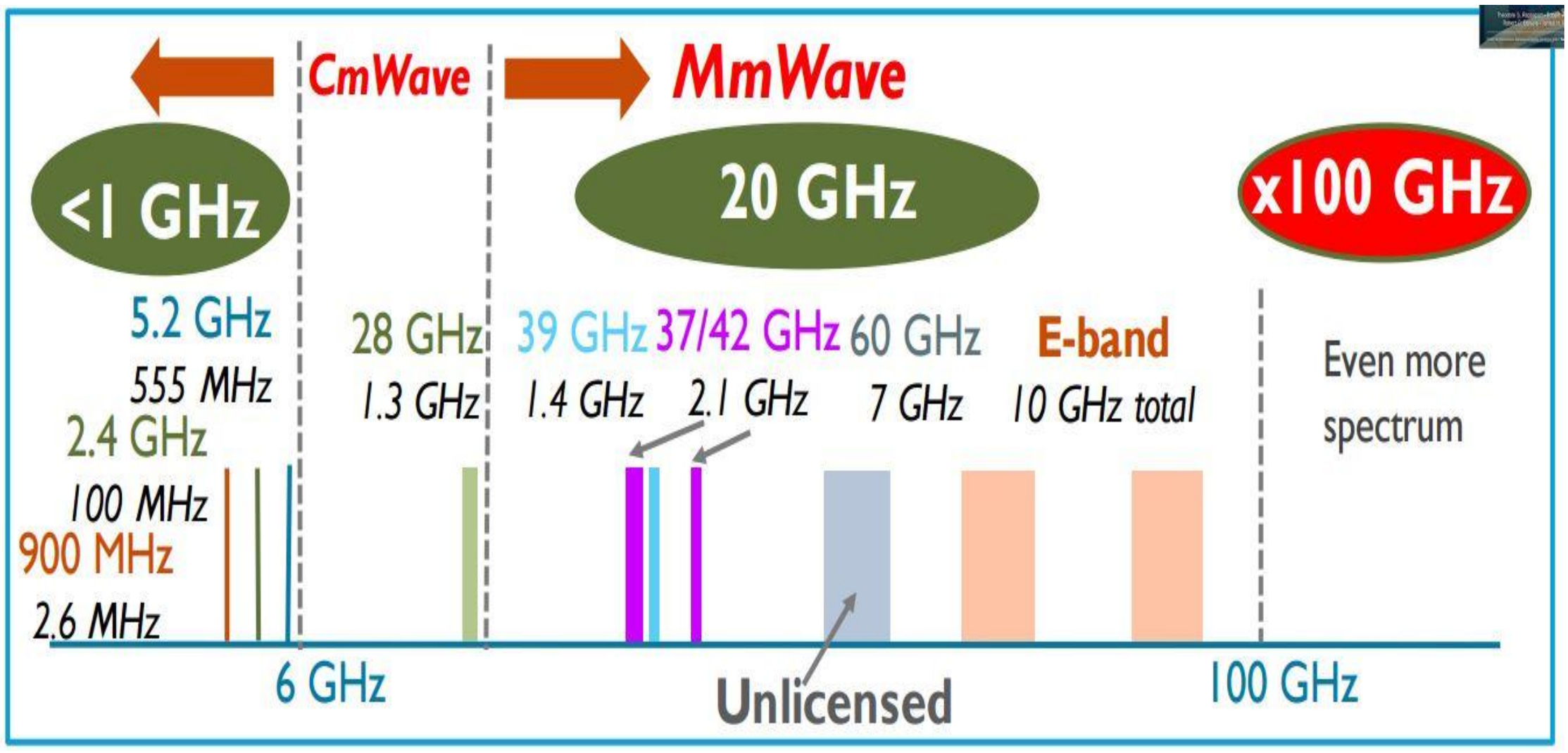




Why called Mmwave?

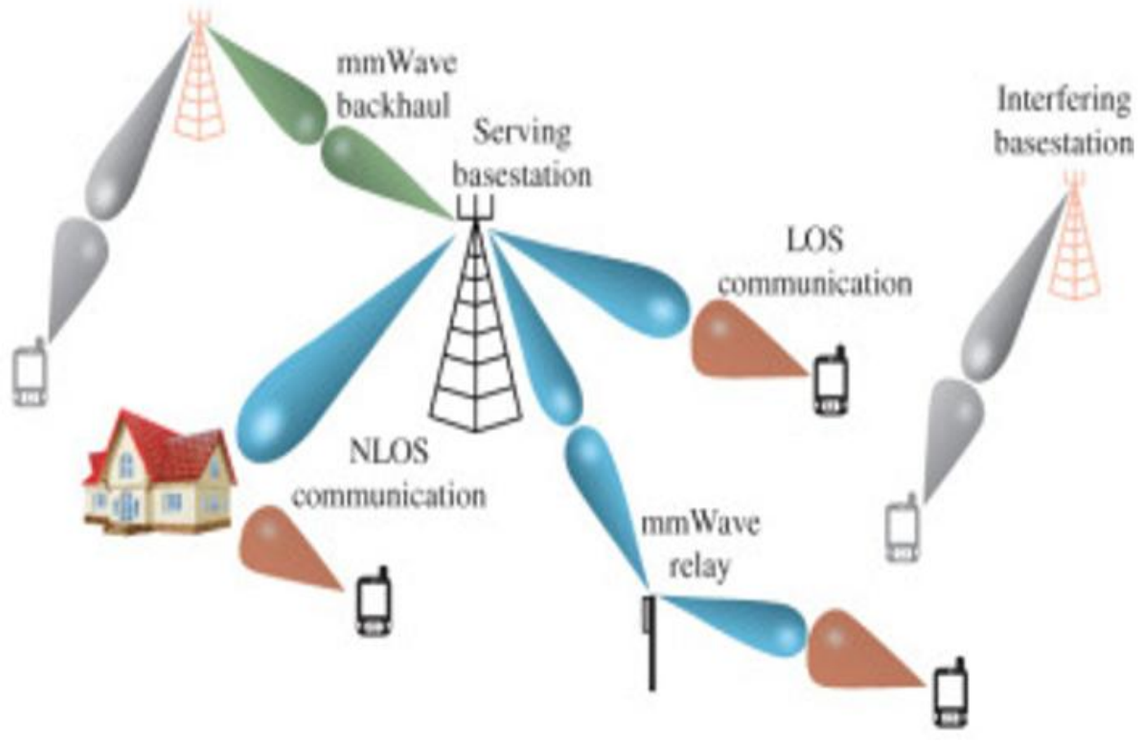
- Frequency between 30GHz-300GHz.
- Wavelength (1-10 mm).
- f=c/λ
- C=3*10^8 m/sec. (speed of light).

Where does it lies?



Millimeter waves (mmwaves) advantages

- The radio spectrum is still rather undeveloped .
- More bandwidth is available.
- Hence higher data rates is achieved.
- Security and privacy is better at mm-wave.



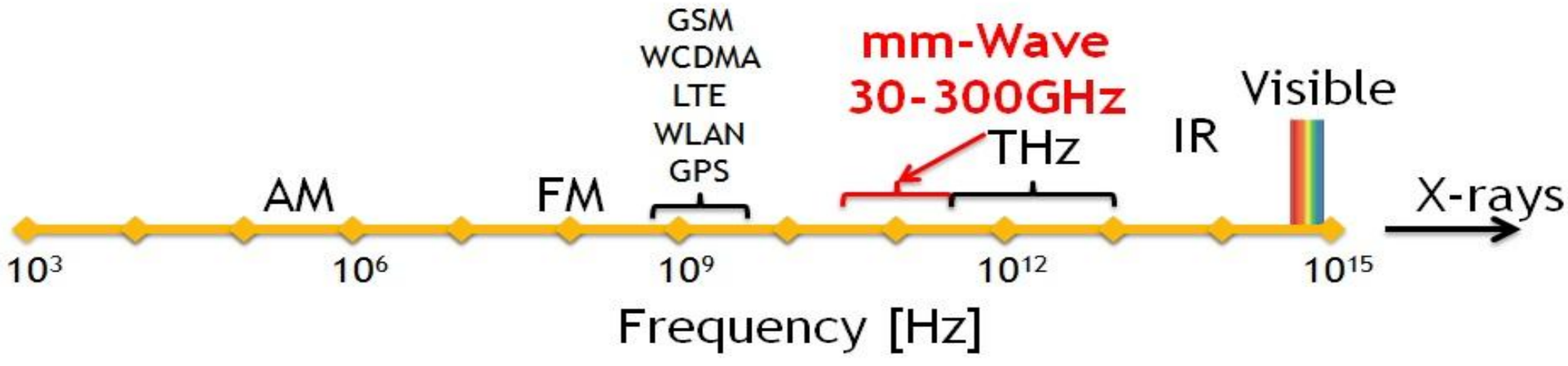
Mmwaves limitations

- Higher costs in manufacturing of greater precision hardware due to components with smaller size.
- At extremely high frequencies, there is significant attenuation. Hence millimeter waves can hardly be used for long distance applications.
- The penetration power of mm-wave through objects such concrete walls is known less.
- There are interferences with oxygen & rain at higher frequencies therefore further research is going on to reduce this.

Mmwaves vs. Microwave vs. Visible light

mm-Wave Signals

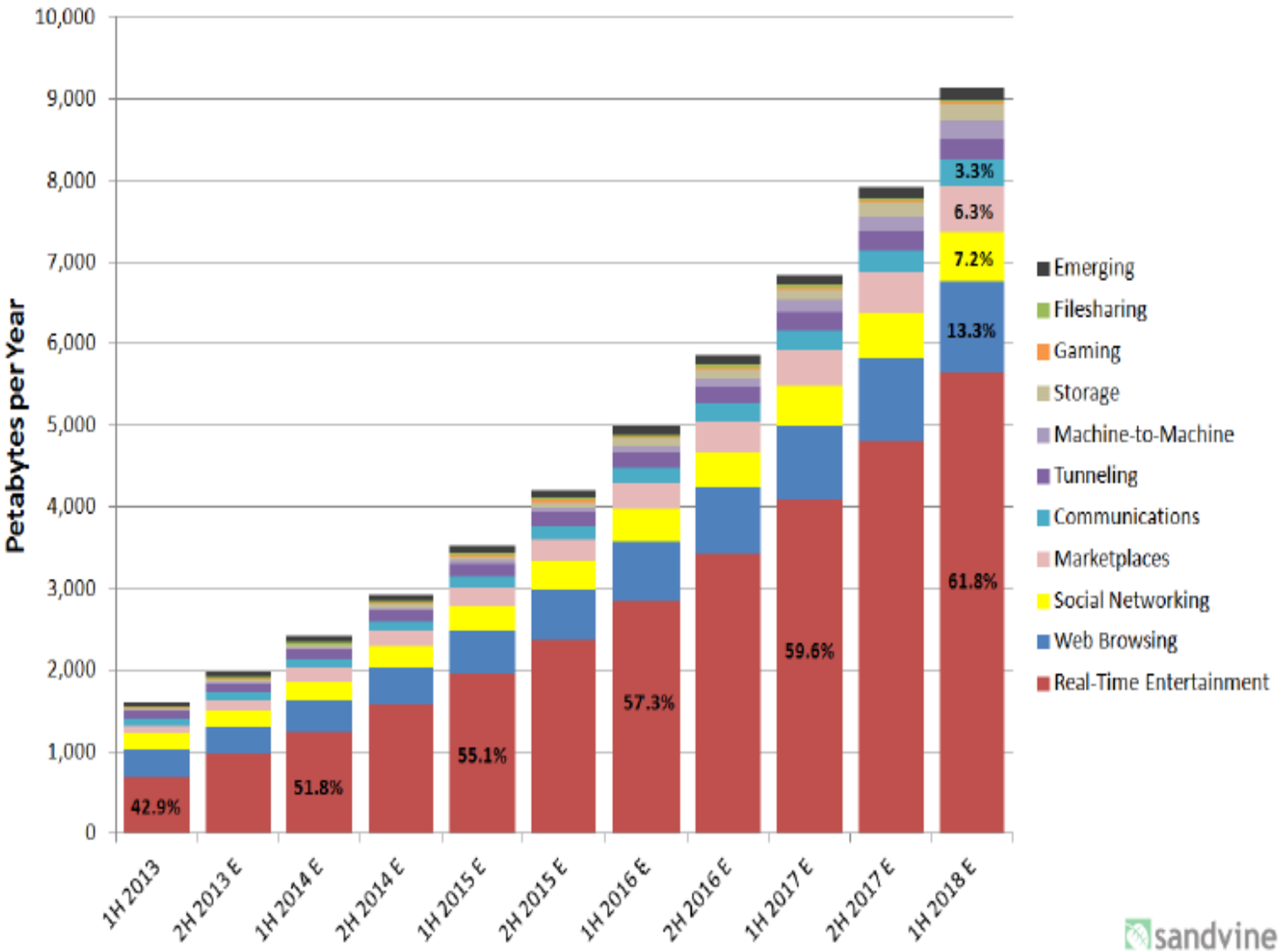
Property	Microwave	mm-Wave	IR/Visible Light
Generation	Electronic	Electronic	Optics
Spatial resolution	~cm - m	~mm - cm	um
Antenna size	PCB	Package	Package
Cost, area, power	Low	High	Low



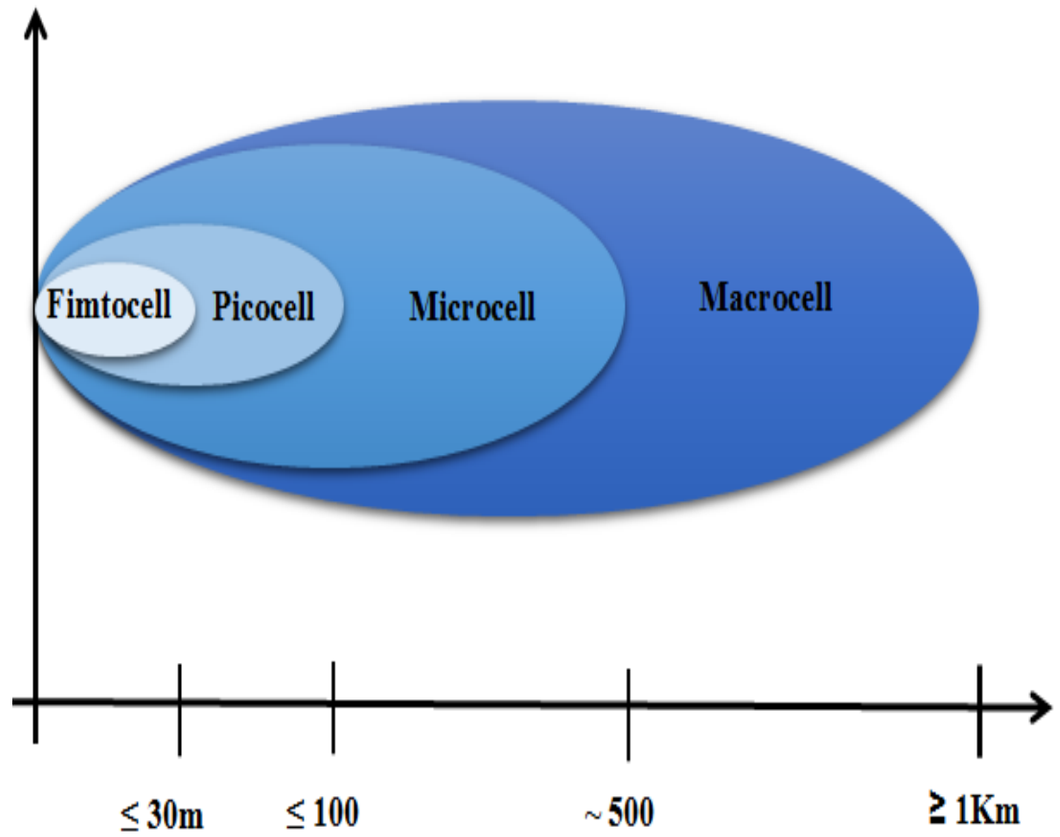
Small cells technology

- In recently studies [**ABI Research]** approximately **80% of data traffic accrues indoor areas**
- Since 5G is on the horizon, so this percentage will increase to 90% in the near future after 5G launch
- Simply this makes the conventional cellular system inadequate and in sufficient to cope with our digital society's demands.
- For increasing system capacity, extending the wireless coverage area, reducing the eNB deployment cost in a high dense residential area, and reducing signaling with carry out the traffic from eNB, **require the small cells technology to overlay on the existing large macro cell called HetNets**

Background



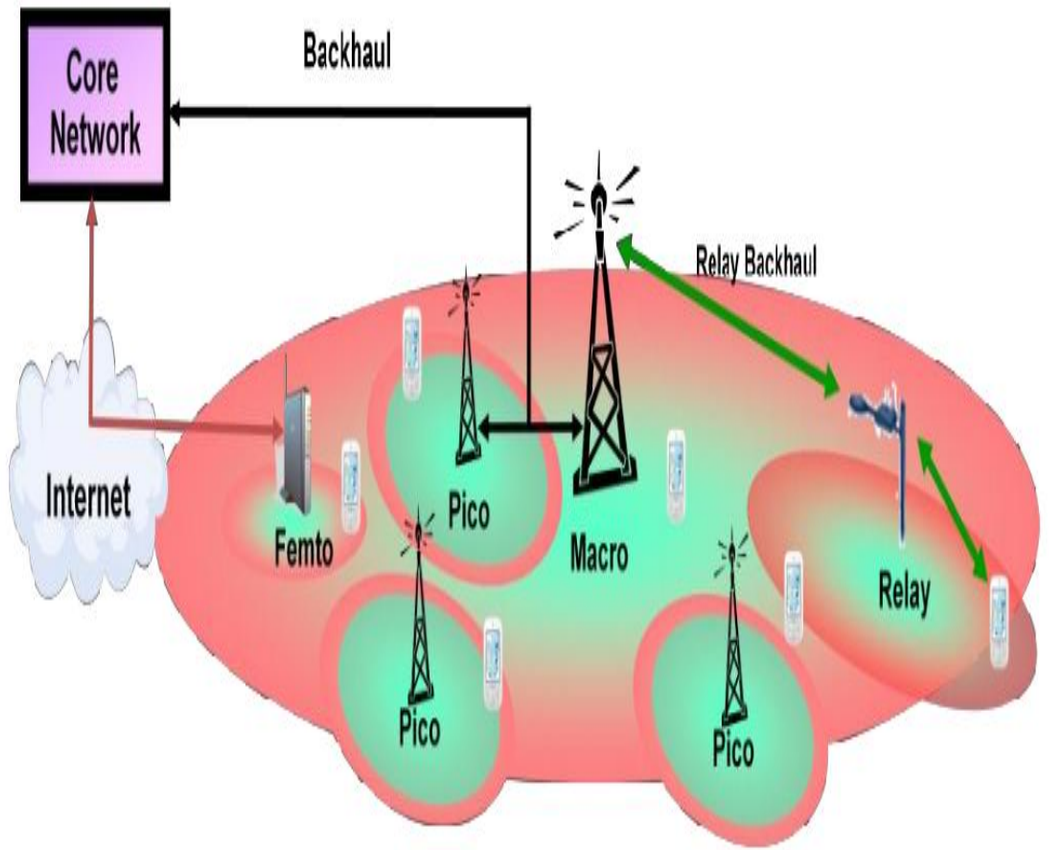
Growth of data traffic in united states[1].



Small cells HetNets comparison in terms of range [5].

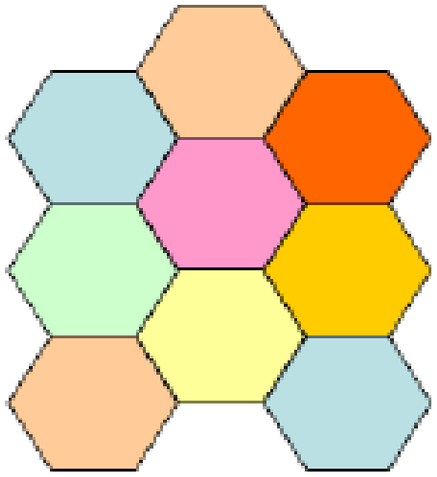
A Comparison Between Low Power Nodes in HetNets

properties	Cell types		
	Micro-cell	Pico-cell	Femto-cell
Power	30dBm	30dBm	20dBm
Coverage range	Up to 500m	<100m	<30m
Backhaul	X2 interface	X2 interface	Home broadband
Access mode	Open to all users	Open to all users	Can be Closed subscriber group
Deployment	Outdoors	Indoors or outdoors	Indoors
Installation	By the operator	By the operator	By the user
Cost	Expensive	Cheap	Very cheap



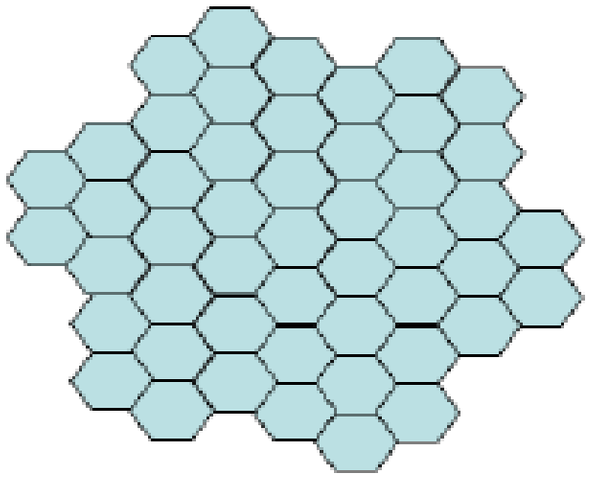
LTE-A Heterogeneous-Networks [15]

Past



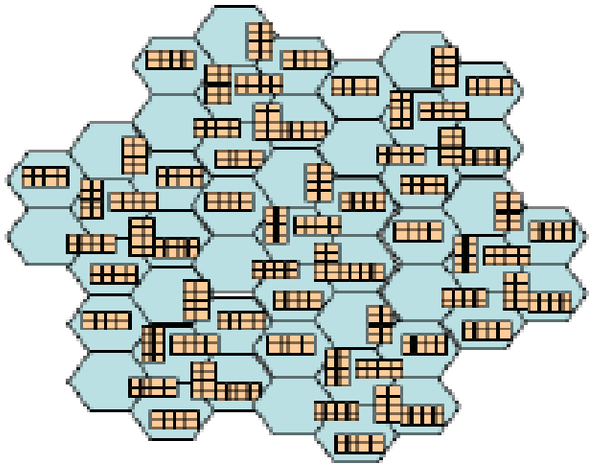
Large macrocells with sparse **planned** reuse and positions

Current



Mainly densified macrocells with **planned** positions and universal reuse

Future



Heterogeneous networks including **uncoordinated** femtocell deployment

Consider the HetNets networks are the best infrastructure and platform to cope with 5G demands for a number of reasons:

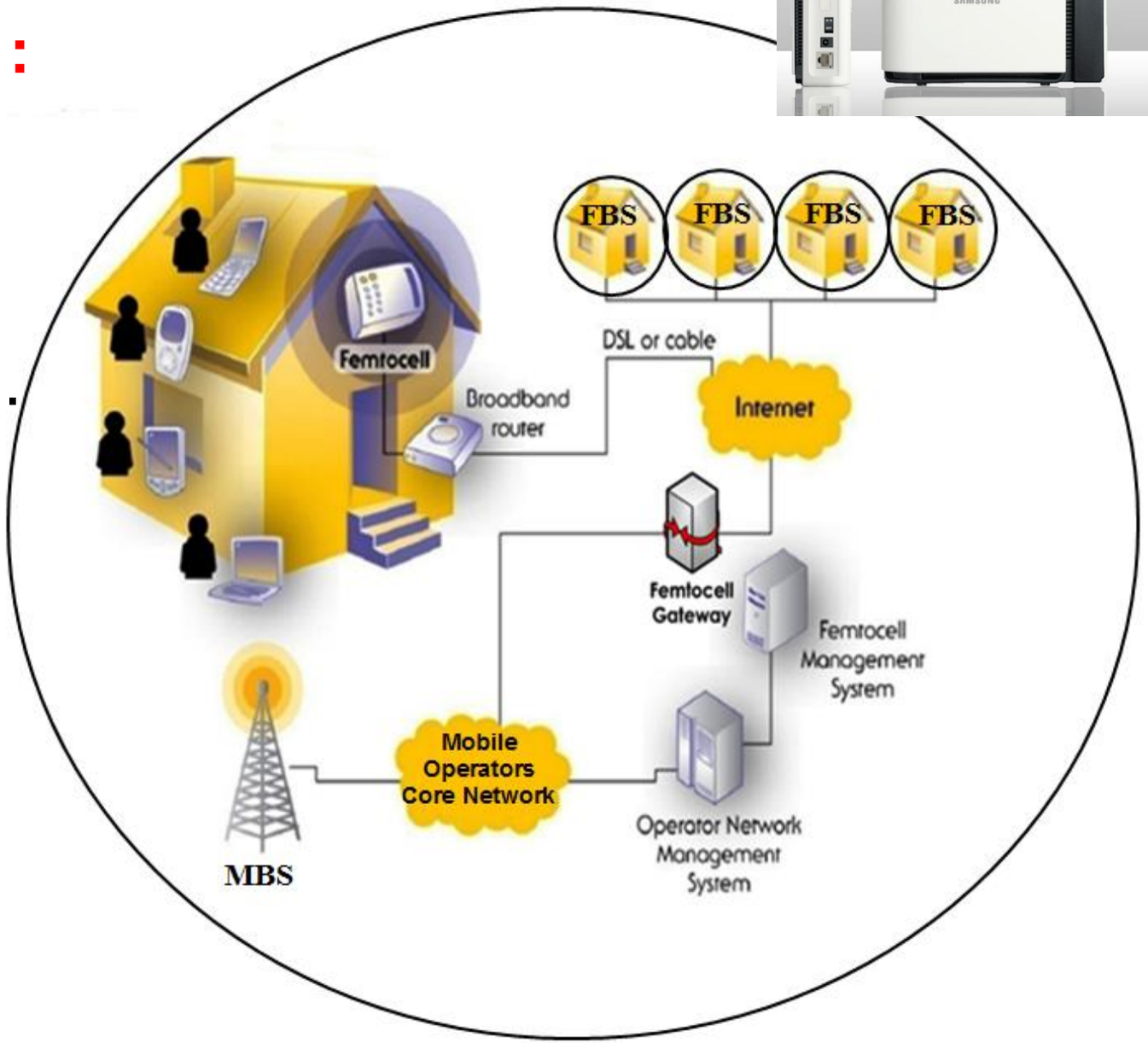
- **First**, and foremost, the **traffic by shrinking** of macro cells size into much smaller cells Where, the large macro base stations may not be able to adequately emerge dense urban areas to serve users in hotspots or dead zones due to many blockages and non-LOS transmissions.
- **A second** aspect is that **most of the high data traffic is produced when the user is at indoor** locations because of the short distance between transmitter and receiver.
- **Third**, a lot of research trying to facilitates trend the **reuse of the resources** where the small cells can reuse the spectrum of macro cells per unit area

Femto cell Technology



the key benefits of the femtocells :

1. **Coverage and capacity improvement**
2. **High data rates and call quality**:.
3. **Improved macrocells reliability**
4. **Cost reduction for mobile operators**
5. **Compelling new femto zone services**
6. **Simple deployment**

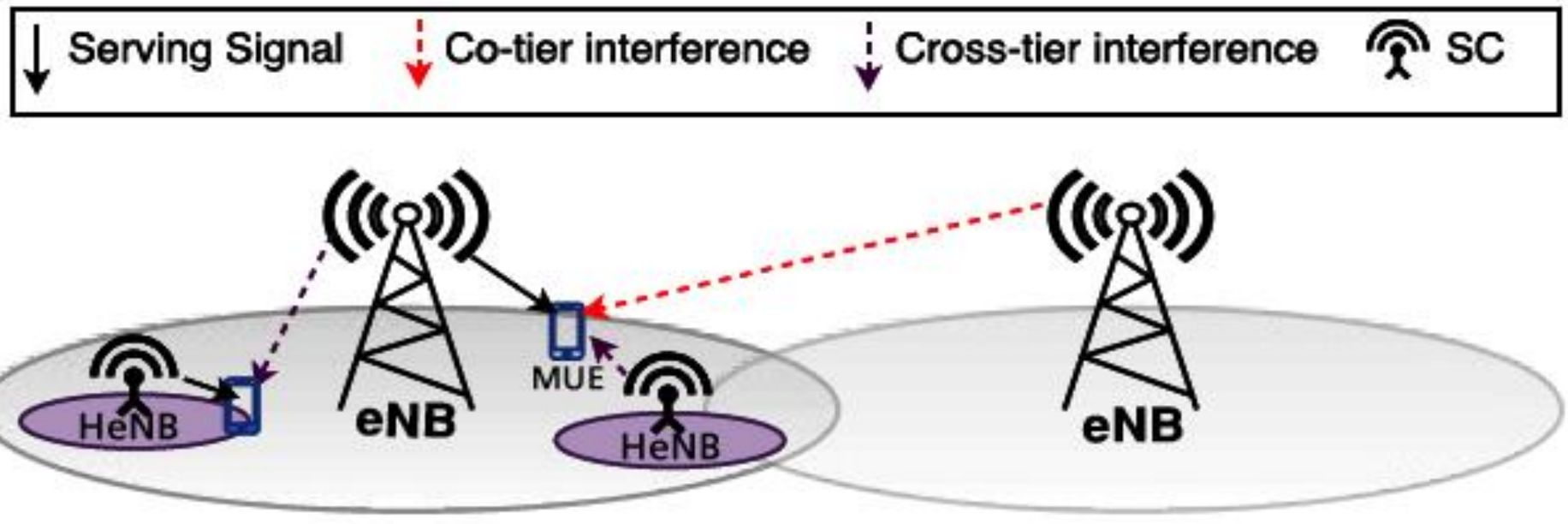


Femto cell challenges

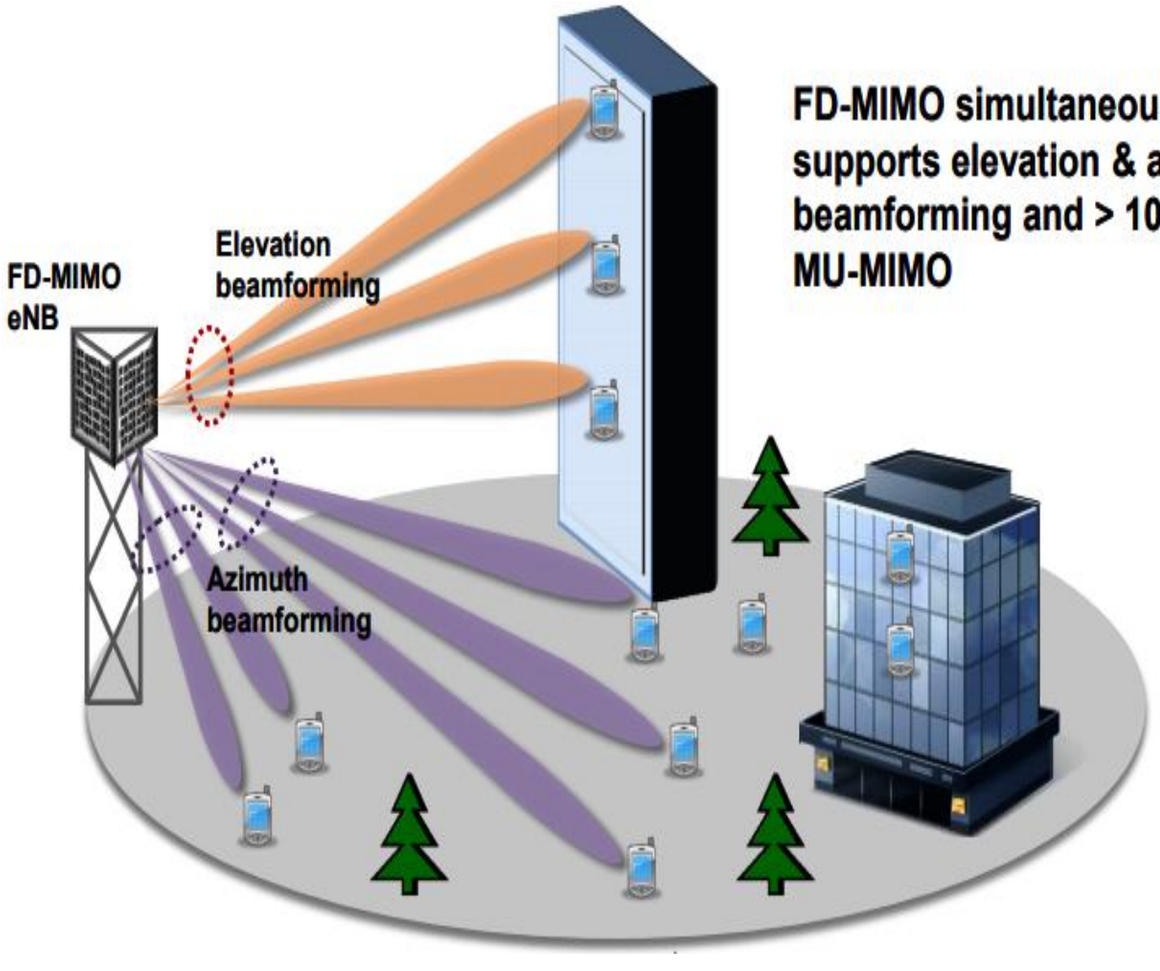
1. How will a femtocell adapt to its surrounding environment and **allocate spectrum** in the presence of intra- and cross-tier interference?
2. How will femtocells provide timing and synchronization?
3. How will backhaul provide acceptable QoS?
4. **How will femtocells handle cross-tier and co tier interference?**
5. **How will femtocells provide access mode (open, closed or hybrid access?**
6. How will handoff be performed in open access?
7. Can subscribers carry their femtocells for use outside the home area?

Interference management and ICI Mitigation

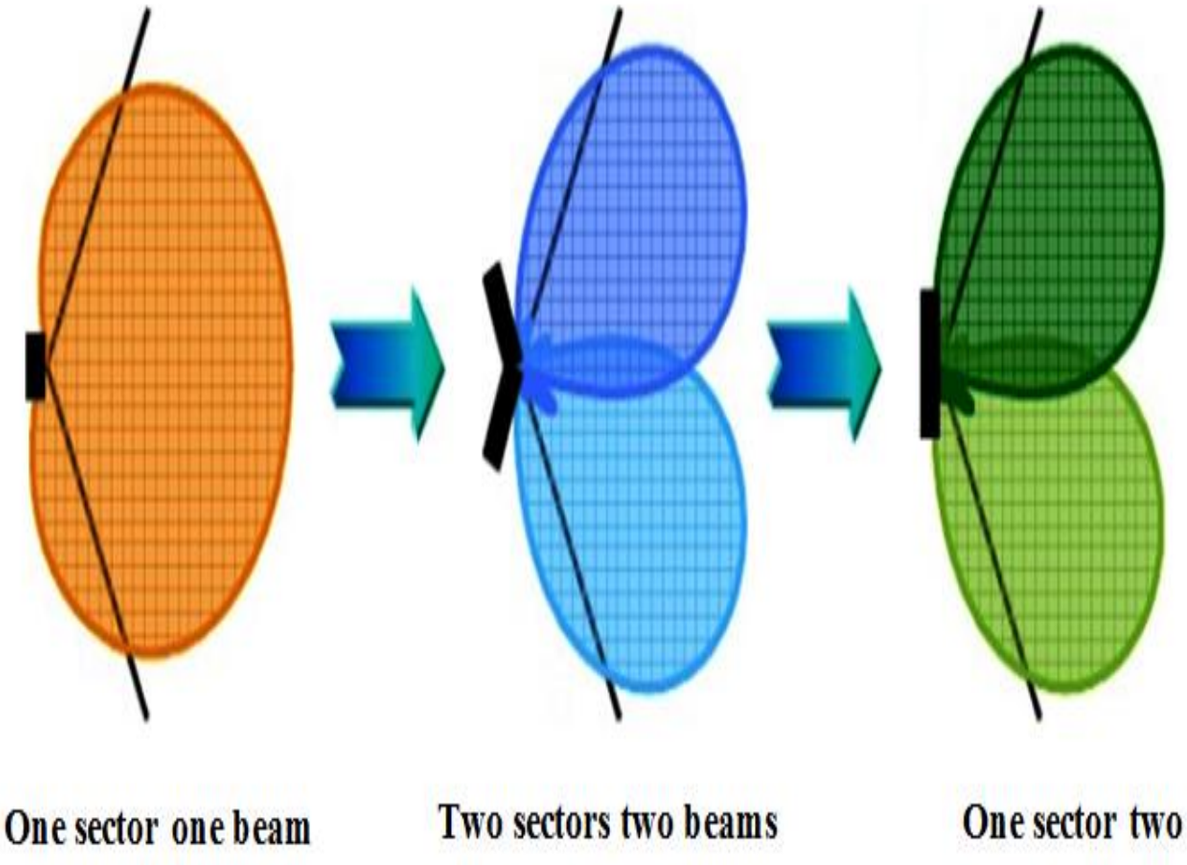
- There are two types of co-channel interference in LTE HetNets are **co-tier and cross-tier interferences** and **cross-tier and cross-tier interferences**



Beamforming Technology



FD-MIMO simultaneously supports elevation & azimuth beamforming and > 10 UEs MU-MIMO

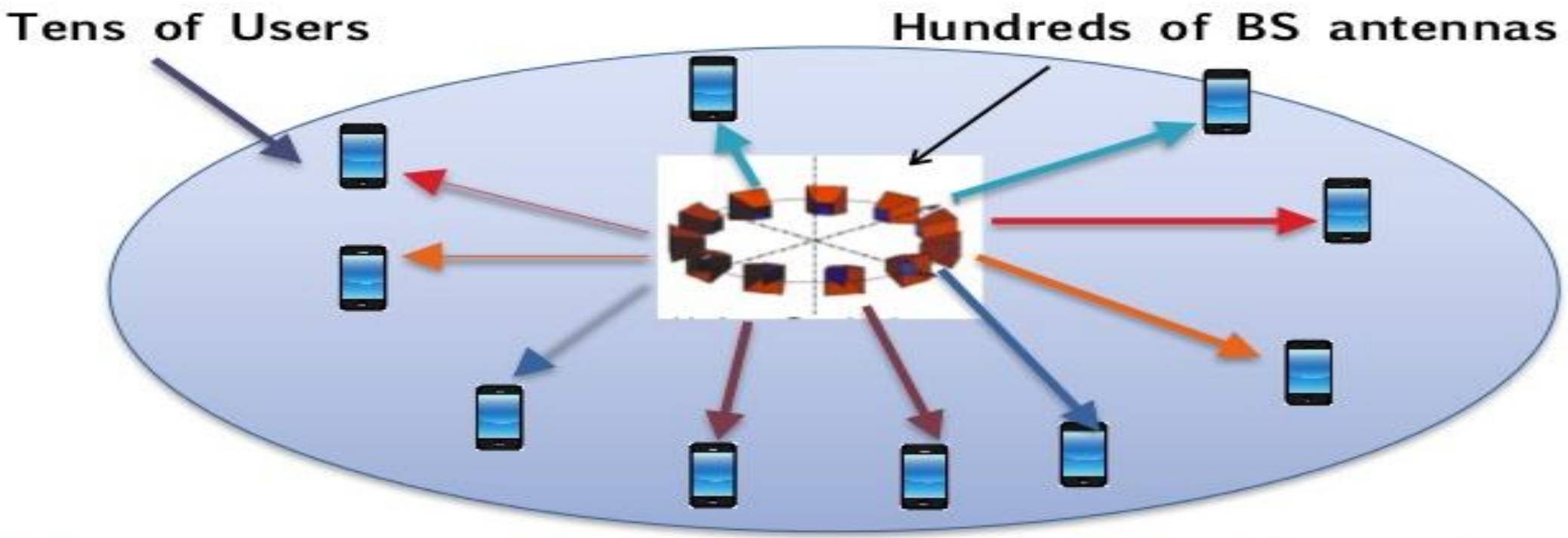


Twin beam technology & FD-MIMO

MASSIVE MIMO **Technology**

What is Massive MIMO

Essentially multiuser MIMO with lots of base station antennas



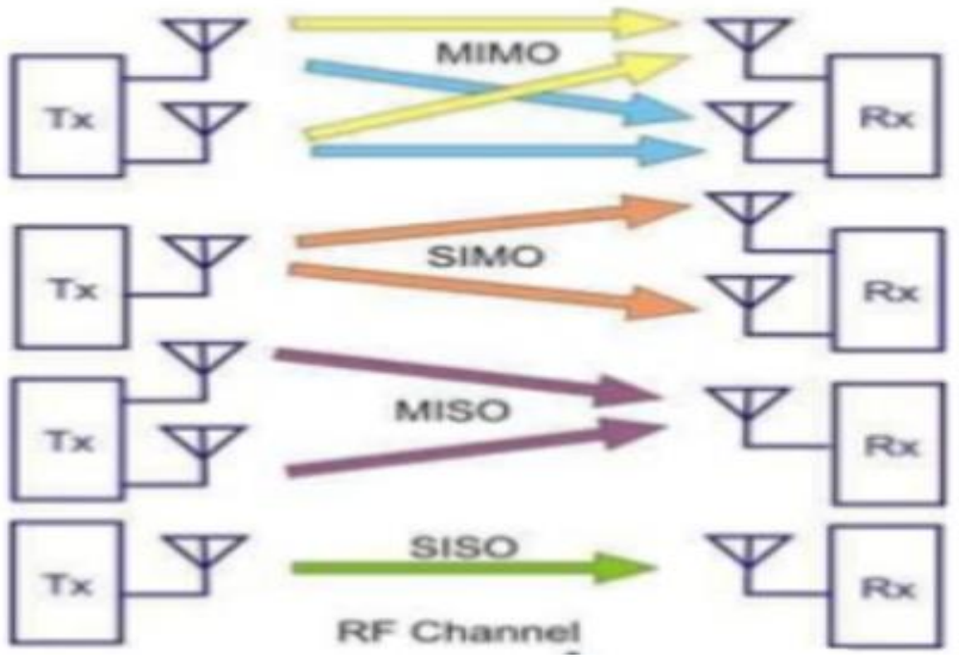
A very large antenna array at each base station
A large number of users are served simultaneously
An excess of base station (BS) antennas

Overview of Massive MIMO

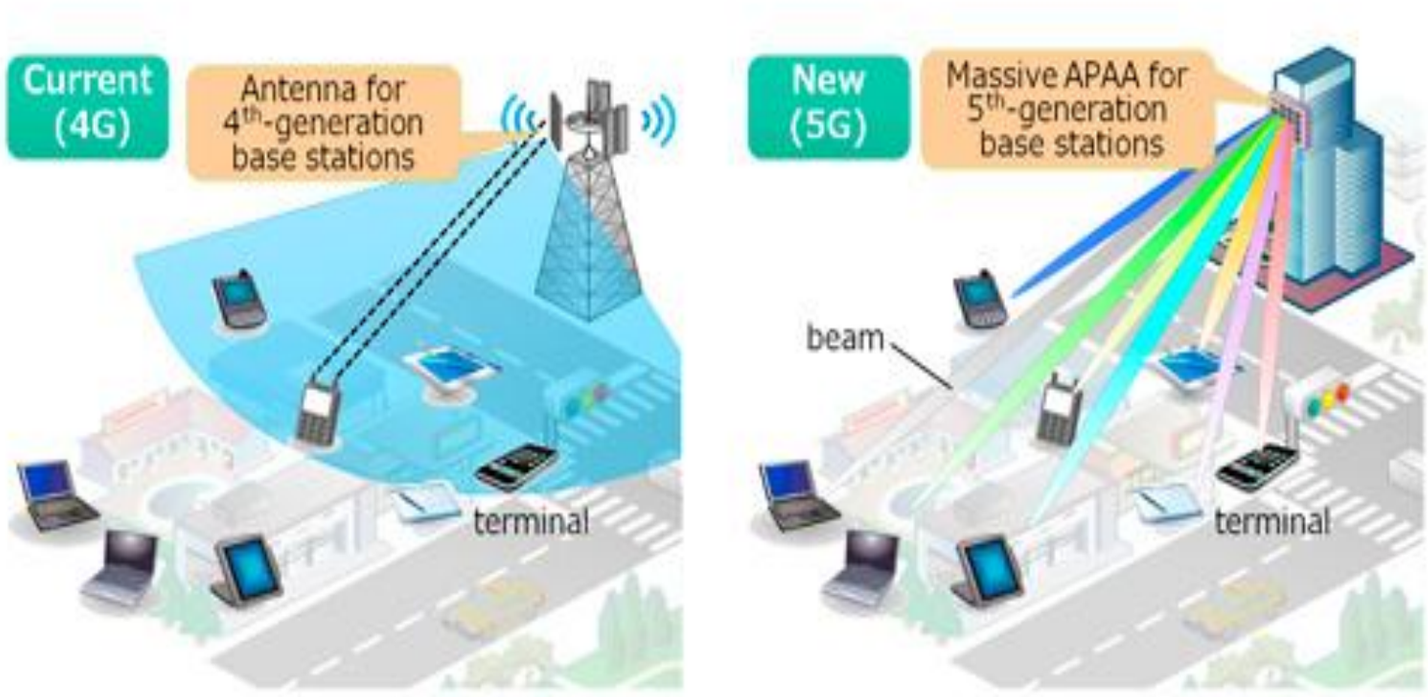
- As the name implies, Massive MIMO is an antenna array system using Massive Amount of Antenna. It is also called Large Scale MIMO.
- Then you may ask "How many Antenna is required to be called 'Massive (Large Scale) MIMO' ?".
- The answer may be different depending on how you design the receiver algorithm (equalizer design).
- We may need over 300 antenna. If you think this is too big number, you may decrease the number by applying a little bit smarter receiver (equalizer) algorithm.
- In conventional LTE using a normal MIMO, the maximum number of antenna in MIMO as of now (Mar 2015) is 8 x 2 or 4 x 4.

MASSIVE MIMO FEATURES:

- Massive (very large no. TX & RX reach more than hundred)
- More signals paths
- High data rate
- Link reliability
- Boost capacity
- More no. RF amplifiers
- More energy consumption
- Complexity increased
- Cost increased



4G Vs. 5G


Current (4G)
Antenna for 4th-generation base stations
terminal
New (5G)
Massive APAA for 5th-generation base stations
beam
terminal

Fifth Generation (5G) systems

- 5G denotes the 5G can be a complete wireless communication without limitation, which bring us perfect real world wireless, supportable to the World Wide Wireless Web (WWWW). According to the Group Special Mobile Association (GSMA) to qualify for a 5G a connection should meet most of these eight criteria:
- 1. One to **10Gbps** connections to end points in the field
- 2. **One millisecond** end-to-end round trip **delay**
- 3. 1000x bandwidth per unit area
- 4. 10 to 100x number of connected devices
- 5. (Perception of) **99.999 percent availability**
- 6. (Perception of) **100 percent coverage**
- 7. 90 percent reduction in network energy usage
- 8. Up to ten-year battery life for low power, machine-type devices

Note: 5G should be rolled out by 2020 to meet business and will meet consumer needs such as:

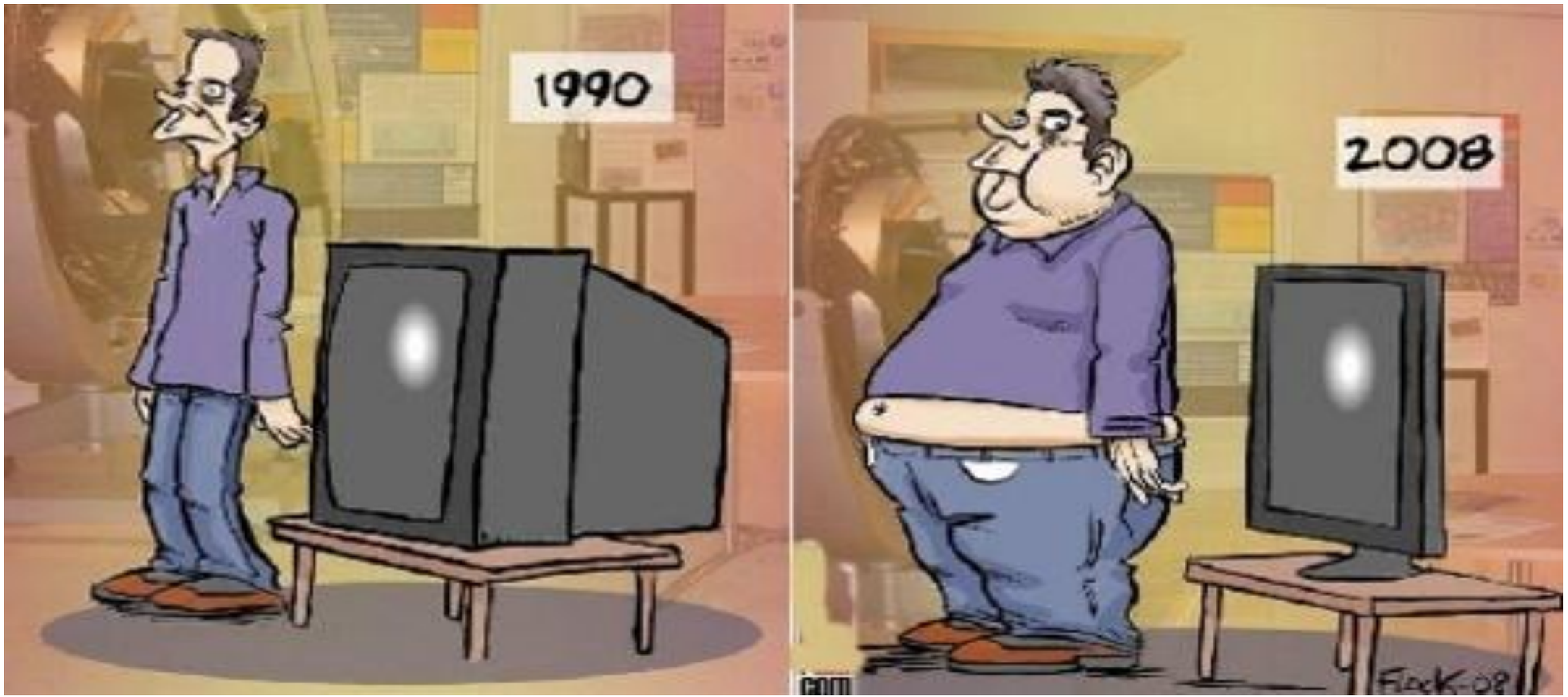
- **Internet of Things (IoT).**
- Broadcast-like services and lifeline communication in times of natural disaster.

Q1/ true OR false

1. **Millimeter wave (mm-Waves), HetNets using small cell technology, Beam-forming technique and Massive MIMO are four keys common inventions that will enable the evolution of cellular networks to support 5G and serve future users demands.**
2. **The lower interference with oxygen and rain as well as the higher penetration of walls and concrete at the The extremely higher frequency are the most difficulties facing Millimeter wave to support 5G.**
3. **In recently studies approximately more than 85% of data traffic accrues indoor areas due to COVID 19 so require the small cells technology to overlay on the existing large macro cell called HetNets.**
4. **Small cells technology are used for reducing the eNB deployment cost in a high dense residential area, and reducing signaling with carry out the traffic from eNB as well as used For increasing system capacity, extending the wireless coverage area.**
5. **The key benefits of the Small cells technology is Prolong the battery life of mobile station.**
6. **Beamforming Technology allows you deliver higher signal quality to your receiver with higher interference.**
7. **The key benefits of the MASSIVE MIMO are High data rate ,Link reliability, Boost capacity due to massive number TX & RX reach more than hundred.**

- Q2/Write short notes about **Millimeter wave**

THANK YOU



ARE … WE…. EVOLUTING???